
3AL Tutorial 7 2024

1. Consider the group $\mathbb{Z}_2$ (under addition) and the set $\mathbb{R}^2 = \mathbb{R} \times \mathbb{R}$. We define a map $*$: $\mathbb{Z}_2 \times \mathbb{R}^2 \rightarrow \mathbb{R}^2$ by

$$[0] * (x, y) = (x, y)$$

$$[1] * (x, y) = (y, x),$$

where $(x, y) \in \mathbb{R}^2$.

- (a) Prove that $*$ is a group action.
- (b) Find all orbits, stabilisers and the fixed set of the group action.
2. Let G be an abelian group and X a set. Suppose $*$ is a group action by G on X .
- (a) Prove that $\circ : (G \times G) \times X \rightarrow X$ defined by $(g_1, g_2) \circ x = g_1^{-1} g_2 * x$ is a group action by $G \times G$ on X .
- (b) Let $x \in X$ and let $S_*(x)$ denote the stabiliser of x under $*$ and $S_\circ(x)$ denote the stabiliser of x under $\circ$. Prove that $(a, b) \in S_\circ(x)$ if and only if $aS_*(x) = bS_*(x)$.
3. Let G be a group. Consider each of the following maps $*_i : G \times G \rightarrow G$. Prove whether each $*_i$ is a group action or not.

(a) $g *_1 x = xg$

(b) $g *_2 x = xg^{-1}$

(c) $g *_3 x = x^{-1}g$

4. Let G be a finite group acting on a set X . Prove that for all $x \in X$, we have $|G| = |S(x)||G * x|$.
5. Let G be a group acting on a finite set X where $|G| = p$ for some prime p with $p \nmid |X|$. Prove that X_f is nonempty.
6. Let G be a group acting on a set X . The **fixer** of a group action is a subset F of G defined by $F = \{g \in G : g * x = x \text{ for all } x \in X\}$.
- (a) Prove that $F \triangleleft G$.

(b) Show that X is a G/F -set.

(c) Prove that the fixer of the action from (b) is the trivial subgroup of G/F .

HINT: Use $*_0$ and F_0 to denote the action by G/F on X and its fixer. This will avoid confusion with the action $*$ by G on X and its fixer F .